

Nouraldin Farge

Software Engineer - Desktop & Local-First Systems

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Summary

Software engineer who shipped three portable Windows applications with public source and verifiable release artifacts, plus a source-free case study of a private system. Owns architecture, trust boundaries, verification, and release decisions for local-first products developed with AI-agent assistance under human review.

Technical Skills

Languages: TypeScript, JavaScript, Rust, Python, SQL, HTML/CSS

Desktop & Frontend: React, Tauri, Electron, Windows, WebView2, Vite

Data & Integration: SQLite, Node.js, Express, REST, WebSockets, JSON Schema, Zod

Quality & Release: Git, GitHub Actions, CI/CD, CodeQL, Vitest, pytest, Playwright, Cargo, Clippy, SBOMs, SHA-256, provenance attestations

Projects

DrawScope v0.6.5 [Code] [Release] | *React, Rust/Tauri, Python, SQLite* 2026-Present

- Built a portable research workbench backed by a reproducible 41,598-row SQLite archive spanning six lottery games, with row-level provenance and documented data gaps.
- Prevented look-ahead bias across 250 walk-forward trials by enforcing a fixed 60/40 discovery-confirmation split with contract tests across TypeScript, Python, and Rust.

GameVault v0.3.5 [Code] [Release] | *React, Rust/Tauri, SQLite* 2026-Present

- Developed a portable game library and launcher with a relative-path SQLite catalog, launch history, backups, and optional metadata sourced only from allowlisted official store pages.
- Hardened review-gated ZIP imports against path traversal, unsafe Windows names, links and reparse points, and decompression abuse; required approval before promotion and supported rollback.

Day-Trading Teacher v0.32.6 [Code] [Release] | *React, Rust/Tauri, local JSON, Zod* 2026-Present

- Built a nine-lesson learning environment that connects decision plans, historical replay, paper practice, journaling, reflection, and spaced review in one local-first workflow.
- Implemented deterministic decimal calculations in Rust, schema-validated lesson imports, and provenance-aware review of Fidelity trade history; deliberately excluded live signals, advice, and order execution.

Research Studio v0.1.0-alpha.21 [Case Study] | *Electron, React, TypeScript, Express, SQLite* 2026-Present

- Architected a private bilingual catalog-enrichment alpha validated against 31,521 series and 2,242,170 episodes, using read-only snapshots, explicit working copies, and batches capped at five records.
- Built schema and evidence gates for untrusted AI output, required human approval before applying changes, and produced recoverable exports; published a source-free case study while keeping the application, data, and build process private.

Experience

Independent Software Engineer 2026-Present

Portfolio Products

- Owned three public applications from requirements through release, including architecture, trust boundaries, automated verification, recovery design, documentation, checksums, SBOMs, and build-provenance attestations.
- Used AI agents for implementation and iteration; personally reviewed changes, resolved failures, validated claims, and retained final authority over safety, licensing, data sources, and releases.

Technical Assistant (Additional Experience) Jun 2015-Sep 2015

Chicago Public Schools

Chicago, IL

- Supported staff and classroom technology, diagnosed hardware and software issues, and explained practical resolutions to non-technical users.

Education

Northeastern Illinois University May 2023

B.A. Computer Science, Cum Laude - GPA: 3.74

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